

Webster University in Tashkent

The Writing Placement Test

This is a 60-minute timed test. Word count: at least 300 words.

You will read a short college-level text and write a response consisting of

1) A Summary followed by 2) An analysis. Directions and questions are found below.

By Gisele Galoustian

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Read the entire article, then respond to the following questions:

Part 1 (Summary) --What is the central idea of the text? Summarize the main points of the author regarding wildfire management strategies in the Western U.S. and their effects on forest health, carbon storage, and climate change.

Part 2 (Analysis) -In your opinion, what is the best way to manage wildfires?

-Should more emphasis be placed on prescribed burns, thinning, or removing dead wood? What are the advantages and disadvantages of these methods?

--How do wildfires affect air quality, human health, and climate change? Should residents be concerned about controlled burns affecting air pollution? How does carbon storage in forests help mitigate climate change?

-How can dead wood be used as a resource instead of being burned? Do you think converting it into biochar is a good solution? Why or why not?

Managing Wildfires in the Western U.S.

For over a century, stopping fires, along with global warming and drought, has led to bigger and more dangerous wildfires in the Western U.S. To prevent this, forest managers use methods like controlled burns, thinning trees, and clearing dry branches. These help reduce fire fuel, protect stronger trees, and keep forests healthy.

However, controlled burns haven't kept up with the fast buildup of dry plants and trees. This has created a "fire deficit"—a gap between how much fuel has piled up and how much is removed. As a result, the risk of extreme wildfires has increased.

Controlled burns help prevent wildfires, but they also have risks. Sometimes, they get out of control and become real wildfires. They also release smoke, making the air unhealthy to breathe. In the Pacific Northwest, smoke from these fires has caused deaths, breathing problems, and work disruptions.

Other things, like logging, drought, and insects, also weaken forests. This makes them less able to absorb carbon, which is important for slowing climate change. Managing wildfires properly can help forests store more carbon and reduce pollution.

For thousands of years, Indigenous Peoples in the Western U.S. have used controlled burns and collected firewood to manage forests. These practices have influenced modern methods, but now scientists are looking for new ways to reduce wildfires—like removing dead wood without burning it.

Researchers from Florida Atlantic University (FAU) studied how taking out dead wood could lower wildfire risks and help forests store more carbon. They tested different methods, such as thinning trees, controlled burns, and removing fallen branches.

The results showed that thinning trees and taking out dead wood reduced wildfire risks and helped store carbon in materials like biochar. Biochar is a type of charcoal that holds carbon and improves soil.

Dr. Scott H. Markwith, a professor at FAU, explained that this method can help forests grow stronger while also cutting down on smoke and pollution. Instead of burning dead wood, turning it into carbon-storing products could be a better solution.

Dr. Rabindra Parajuli, the study's lead author, noted that controlled burns can release a lot of carbon over time. However, removing and turning dead wood into biochar helps reduce emissions and allows forests to store more carbon.

More research is needed to see how this method works over time and in different forests. Long-term studies will help decide if removing dead wood can restore natural wildfire cycles, improve air quality, and make forests in the Western U.S. stronger.